LECTURE 8:
BAYESIAN MODEL SELECTION

Your task is to model the number of terms members of Congress serve. To make things simple, consider a single Congressional district and suppose we observe a sequence of terms lengths $T_1, T_2, \ldots$ where each T_t is a positive integer. We will make the following assumptions:

- The district contains D Democrats and R Republicans (and no independents). This does not change over time. $D+R$ is odd.
- When there is no incumbent, the probability that a voter votes for their party's candidates is p (independent of how they voted in the previous election). When there is an incumbent running, there is an incumbency advantage: among members of the opposition (non-incumbent) party, the probability of voting for their party's candidate is $q < p$ (also independent of past voting).
- Members of Congress continue always run for reelection unless they die or retire. The probability dying/retiring at each election is r (and this process is memoryless).

Please answer the following questions:

1. What is the probability that Democrats win the first election?
2. If the current incumbent is a Democrat, what is the probability that he or she will be reelected? Answer the same question for a Republican.
3. What is the expected number of terms that a Democrat serves? A Republican?
4. In the long run, what proportion of elections are won by Democrats? By Republicans? Justify your answer.
5. What is the size of the incumbency advantage? Discuss.

Key concepts:

- Bayes' Factor
- Relationship between posterior probability of one-side hypothesis and frequentist p-values
- Deviance
- Deviance Information Criterion

1. Introduction

We review briefly posterior predictive checks, hypothesis testing using Bayes' Factor, and model selection using the Deviance Information Criterion (DIC).

2. Posterior Predictive Checks

If $y|\theta \sim p(y|\theta)$ andd $p(\theta)$ is our prior, the marginal distribution of y is

$$p(y) = \int p(y,\theta)\, d\theta = \int p(y|\theta)\, p(\theta)\, d\theta.$$

This is called the *prior predictive distribution*.

Once we have observed some data y and we wish to make an out-of-sample prediction for $\tilde{y}$ generated by the same process (but independent of y conditional upon θ), we have the posterior predictive distribution

$$p(\tilde{y}|y) = \int p(y, \theta|y)\, d\theta = \int p(y|\theta)\, p(\theta|y)\, d\theta.$$

Given a sample from either the prior or the posterior, it's easy to draw a sample from either the prior or posterior predictive distribution. It is considerably more difficult to obtain the full density, though approximations are available in some cases.

We can check the validity of a model by computing test statistics on the sample observations and comparing them to their probability under the posterior predictive distribution. The choice of test statistic is *ad hoc* (as in classical statistics). These are called *posterior predictive checks* or (sometimes) *Bayesian* p*-values*.

3. Bayesian Hypothesis Testing

3.1. **One-sided hypothesis test.** Suppose we have the model $p(y|\theta)$ where $\theta \in \Theta$. Consider the hypotheses

$$H_0 : \theta \in \Theta_0 \quad \textit{vs.} \quad H_1 : \theta \in \Theta_1$$

where $\Theta_0 \cap \Theta_1 = \emptyset$ and $\Theta_0 \cup \Theta_1 = \Theta$. Let the prior be $p(\theta)$, so the prior probabilities of the hypotheses are

$$p(H_0) = \int_{\Theta_0} p(\theta)\, d\theta \qquad p(H_1) = \int_{\Theta_1} p(\theta)\, d\theta$$

The posterior distribution of θ is

$$p(\theta|y) \propto p(\theta)\, p(y|\theta)$$

so the posterior probabilities of the hypotheses are

$$p(H_0|y) = \int_{\Theta_0} p(\theta|y)\, dy \qquad p(H_1|y) = \int_{\Theta_1} p(\theta|y)\, dy$$

We define the *Bayes Factor* to be the ratio of the posterior odds to the prior odds,

$$\text{BF} = \frac{P(H_0|y)/P(H_1|y)}{P(H_0)/P(H_1)}$$

(*BDA3*, p. 182, defines the Bayes' Factor as the likelihood ratio. This definition is identical when the prior probabilities of the model are the same.) Note that

$$\begin{aligned}
\frac{P(H_0)\,\text{BF}}{P(H_0)\,\text{BF} + 1 - P(H_0)} &= \frac{P(H_0)\dfrac{P(H_0|y)/P(H_1)}{P(H_0)/P(H_1)}}{P(H_0)\dfrac{P(H_0|y)/P(H_1|y)}{P(H_0)/P(H_1)} + P(H_1|y)} \\
&= \frac{P(H_1)\,P(H_0|y)}{P(H_1)\,P(H_0|y) + P(H_1)P(H_1|y)} \\
&= \frac{P(H_0|y)}{P(H_0|y) + P(H_1|y)} = P(H_0|y).
\end{aligned}$$

Example 1. *Suppose* $y_1, \dots, y_n \overset{iid}{\sim}$ *Normal*(θ, σ^2) *where* σ^2 *is known. Suppose we have the prior* $\theta \sim$ *Normal*(θ_0, τ_0^2) *and we wish to test*

$$H_0 : \theta \leq \theta_0 \quad \text{vs.} \quad H_1 : \theta > \theta_0$$

Since $P(H_0) = P(H_1) = 1/2$*, the prior odds are equal to one. Since* $\theta|y \sim$ *Normal*(θ_1, τ_1^2)*, where*

$$\theta_1 = \frac{\dfrac{\theta_0}{\tau_0^2} + \dfrac{n\bar{y}}{\sigma^2}}{\dfrac{1}{\tau_0^2} + \dfrac{n}{\sigma^2}} = \frac{\sigma^2\theta_0 + n\tau_0^2\bar{y}}{\sigma^2 + n\tau_0^2}$$

$$\tau_1^2 = \left(\frac{1}{\tau_0^2} + \frac{n}{\sigma^2}\right)^{-1} = \frac{\tau_0^2\,\sigma^2}{\sigma^2 + n\tau_0^2},$$

the posterior probability of H_0 *is*

$$P(H_0|y) = P(\theta \leq \theta_0|y) = \Phi\left(\frac{\theta_0 - \theta_1}{\tau_1}\right)$$

Using

$$\theta_0 - \theta_1 = \theta_0 - \frac{\sigma^2\theta_0 + n\tau_0^2\bar{y}}{\sigma^2 + n\tau_0^2} = \frac{n\tau_0^2(\theta_0 - \bar{y})}{\sigma^2 + n\tau_0^2} = \frac{n\tau_1^2(\theta_0 - \bar{y})}{\sigma^2},$$

we obtain

$$P(H_0|y) = \Phi\left(\frac{n\tau_1(\theta_0 - \bar{y})}{\sigma^2}\right) = \Phi\left(\sqrt{\frac{n\tau_0^2}{\sigma^2 + n\tau_0^2}}\left(\frac{\theta_0 - \bar{y}}{\sigma/\sqrt{n}}\right)\right)$$

For large τ_0^2*, this is approximately equal to the frequentist* p*-value* $\Phi(\sqrt{n}(\theta_0 - \bar{y})/\sigma)$*. But, for informative priors, the two can differ substantially.*

3.2. **Summary of Results for Two-sided Hypotheses.** The approach above does not extend particularly well to two-sided hypotheses. The point null $H_0 : \theta = \theta_0$ has zero prior probability. We can attempt to solve this by *assigning* $P(\theta = \theta_0) = p_0 > 0$, so $H_1 : \theta \neq \theta_0$ has prior probability $P(H_1) = 1 - p_0$.

In this case, the prior for θ is a mixture of $P(\theta = \theta_0|H_0) = 1$ and

$$p(\theta|H_1) = (1 - p_0)(2\pi)^{-1/2}(\tau_0^2)^{-1/2}\exp\left\{-\frac{(\theta - \theta_0)^2}{2\tau_0^2}\right\}$$

We get some cancellation when computing the the Bayes Factor:

$$\text{BF} = \frac{p_0\dfrac{\sqrt{n}}{\sigma}\exp\left\{-\dfrac{n}{2\sigma^2}(\bar{y} - \theta_0)^2\right\}}{(1 - p_0)(\sigma^2/n + \tau_0^2)^{-1/2}\exp\left\{-\dfrac{(\bar{y} - \theta_0)^2}{2(\sigma^2/n + \tau_0^2)}\right\}}$$

However, the Bayes' Factor is quite sensitive to the variance of the prior. As $\tau_0^2 \to \infty$, the posterior probability of H_0 tends to one.

4. Predictive Accuracy

Point prediction: a single point estimate is made, usually the expected value given the model, and fit is measured using mean square error.

Probabilistic prediction: usually evaluated using log predictive density (a.k.a. the log likelihood).

Log predictive density is proportional to the MSE if the model is normal with constant variance.

Notation:

$f(y)$ the data generating process for both observed and future data.

$p(y|\theta)$ and $p(\theta)$ specified model and prior

$p_{\text{post}}(\theta)$ posterior based on data y, model $p(y|\theta)$, and prior θ.

$p_{\text{post}}(\tilde{y}_i) = \mathbb{E}[p_{\text{post}}(\tilde{y}_i|\theta)] = \int p(\tilde{y}_i|\theta)\, p_{\text{post}}(\theta)\, d\theta$ posterior predictive distribution.

$\log p_{\text{post}}(\tilde{y}_i)$ log posterior predictive distribution for a single future observation

$\mathbb{E}_f[\log p_{\text{post}}(\tilde{y}_i)] = \int [\log p_{\text{post}}(\tilde{y}_i)]\, f(\tilde{y}_i)\, dy_i$ Expected Log Posterior Distribution (elpd)

elppd (pointwise elpd) $\sum_{i=1}^n \mathbb{E}_f[\log p_{\text{post}}(\tilde{y}_i)]$

Neither of these is feasible to compute since f is unknown.

5. Information Criteria

In classical statistics, we test nested models $M_0 \subset M_1$ using the *likelihood ratio statistsic*,

$$\text{LR}\frac{p(y|\hat{\theta}_0)}{p(y|\hat{\theta}_1)}$$

where $\hat{\theta}_1$ is the (unrestricted) MLE and $\hat{\theta}_0$ is the restricted MLE:

$$\hat{\theta}_1 = \text{argmax}_{\theta\in\Theta_1} p(y|\theta) \qquad \hat{\theta}_0 = \text{argmax}_{\theta\in\Theta_0} p(y|\theta)$$

where $\Theta_0 \subset \Theta_1$, since the models are nested. Under mild regularity conditions

$$-2\log \text{LR} \xrightarrow{P_0} \chi^2(k_1 - k_0)$$

where k_i is the number of parameters in model M_i (*i.e.*, the dimension of Θ_i).

Claim: Even if $\theta \sim p(\theta)$ and $p(y|\theta)$ is correctly specified, $\mathbb{E}_{\text{post}}[\log p(\tilde{y}|\hat{\theta})] > \mathbb{E}_f[\log p(\tilde{y}|\theta)]$.

Naive test: estimate out-of-sample $\mathbb{E}[\log p(\tilde{y}|y)]$ using $\log p(\tilde{y}|\hat{\theta})$. Equivalently, we can use the deviance,

$$D = -2\log p(y|\hat{\theta})$$

Data is being used twice and overestimates fit. This can be corrected by adding an additional term

$$\text{DIC} = D - p_D$$

where p_D is a penalty for the number of parameters estimated.

Solution: correct for effective number of parameters.

$$p_{\text{DIC}} = 2\left(\log p(y|\hat{\theta}) - \mathbb{E}_{\text{post}}[\log p(y|\theta)]\right)$$

where $\hat{\theta}$ is the posterior mean. The expectation (called the "penalty" in JAGS) is estimated by simulation.

Then we calculate the expected log predictive distribution using

$$\log p(y|\hat{\theta}) - p_{\text{DIC}}$$

This is implemented in both JAGS and Stan.